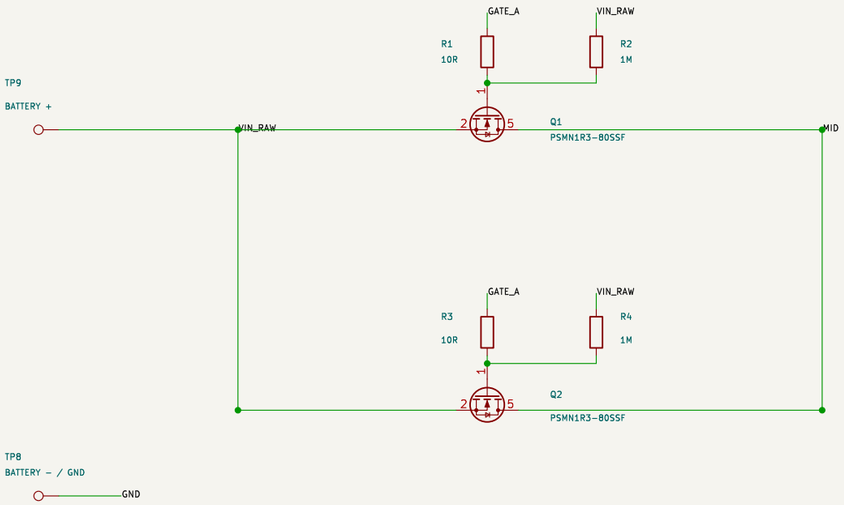
A. MAIN POWER PATH — 60 A TARGET, LAYOUT / THERMAL QUALIFICATION PENDING

6S: 25.2 V maximum. This is a reverse-polarity stage, not a short-circuit disconnect.

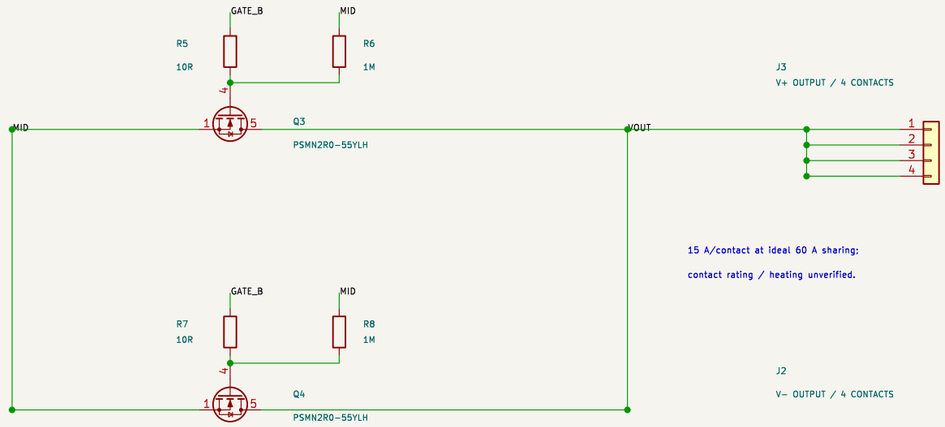
UPSTREAM: full-pack reverse tolerance

Two parallel 80 V MOSFETs



DOWNSTREAM: NCV68261 RPP mode

Two parallel 55 V logic-level MOSFETs



DRAFT: static reverse-polarity architecture.
Forward body diodes bypass an OFF gate.

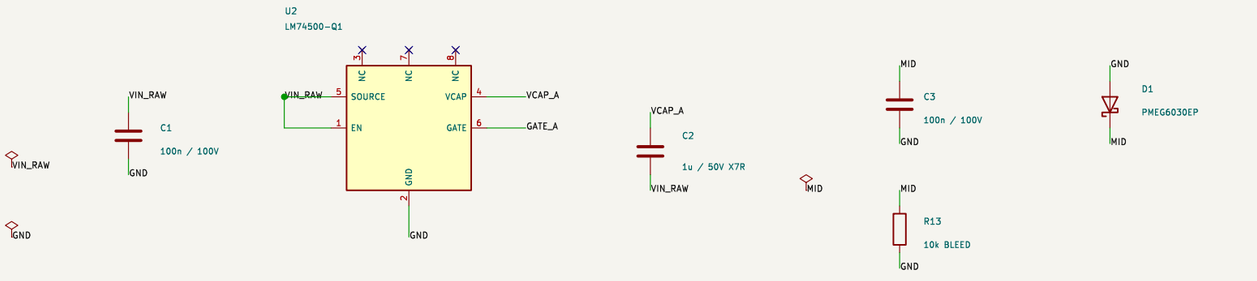
No positive-overvoltage isolation,
short-circuit limiting or thermal shutdown.

Check hot reversal, startup SOA,
gate drive and 60 A thermal limits.

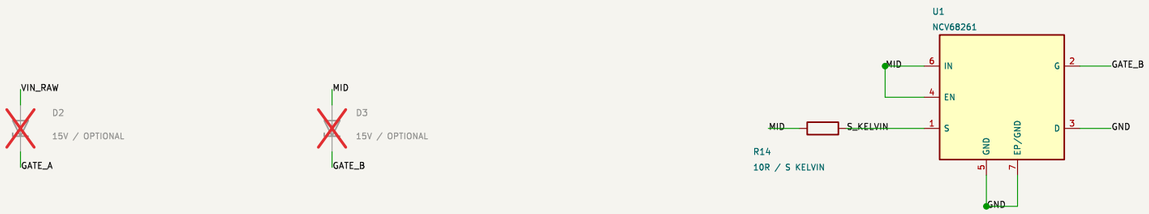
NCV 511DW land envelope checked.
2 oz mask/stencil approval pending.

B. CONTROLLERS / DECOUPLING / CLAMPS

D1: negative MID clamp: pulse energy/current must be verified.



C2: effective capacitance >= 333 nF at operating bias.



C. BALANCE INTERFACE — PROTECTION NOT IMPLEMENTED

TP1..TP7 are raw cell taps; J1 is the existing balance output. All 14 endpoints are intentionally open.



No pass-through jumpers. BMS limits and balance-current rating are required before connecting these paths.



60 A target only; current and fault qualification pending
Balance protection not implemented; interfaces left open
Engineering evaluation

Sheet: /
File: ReversePolarityProtection4MOSFET.kicad_sch

Title: 25 V / 60 A target — four MOSFET evaluation

Size: A3	Date: 2026-10-15	**Rev: C**
KiCad E.D.A. 9.0.7		Id: 1/1